

Yu Zhang

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Educational Background

University of Galway <i>Ph.D. in Computer Science</i>	Galway, Ireland Sept 2025 – Sept 2029 (expected)
University of Galway <i>M.Sc. in Computer Science – Artificial Intelligence (First Class Honours)</i>	Galway, Ireland Sept 2024 – Sept 2025
Heilongjiang University of Science and Technology <i>B.Eng. in Computer Science and Technology (Major Degree)</i> GPA: 3.55 (85.5%) <i>B.Sc. in Mathematics and Applied Mathematics (Minor Degree)</i> GPA: 3.75 (87.5%)	Harbin, China Oct 2020 – Jun 2024

Research Interests

Reinforcement Learning, Robotics

Research Paper

- **Yu Zhang**, Karl Mason. “D-SPEAR: Dual-Stream Prioritized Experience Adaptive Replay for Stable Reinforcement Learning in Robotic Manipulation.” *arXiv:2603.27346*, 2026. (Accepted at ICCRE 2026), [\[Link\]](#)

Research & Project Experience

D-SPEAR: Dual-Stream Prioritized Experience Adaptive Replay for Stable Reinforcement Learning in Robotic Manipulation Oct 2025 – Mar 2026

GitHub: <https://github.com/CorradoZhang/D-SPEAR-Code-and-Result>

- Designed and implemented D-SPEAR, a Soft Actor-Critic based reinforcement learning algorithm with a dual-stream prioritized replay mechanism, emphasizing high-TD samples for the critic, low-TD samples for the actor, and a shared anchor subset to stabilize updates across both branches.
- Implemented an adaptive replay coefficient λ to adjust the balance between uniform and prioritized sampling based on the coefficient of variation of TD errors; additionally integrated candidate-ratio sampling and Huber critic loss to improve robustness and reward-scale adaptability.
- Built a robotic manipulation benchmark pipeline with robosuite and conducted 5-seed experiments on Lift and Door; measured by the mean return over the last 10 episodes, D-SPEAR achieved 305.4 on Lift, outperforming the best baseline TD3 (164.9) by 85.3% and SAC (156.9) by 94.7%.
- Completed the full research workflow including ablation studies, parallel multi-seed training, result aggregation, and publication-ready visualization; on Lift, removing the dual-stream design, low-TD actor sampling, or high-TD critic sampling reduced the mean return to 199.9, 179.0, and 184.2, corresponding to drops of about 34.6%, 41.4%, and 39.7% from the full model (305.4), validating the effectiveness of the core design.

Design of Crack Detection System for USTH Road Based on Image Processing Harbin, China
Graduation Capstone Project for B.Eng. in Computer Science and Technology, Score: 93/100 (Rank: 1/158)

- Developed a Python-based road crack image collection system, enabling automated image capture both inside and outside the Heilongjiang University of Science and Technology campus.
- Designed an image processing system in MATLAB to process and display collected images, incorporating techniques such as histogram equalization and binary filtering.
- Implemented automatic crack severity classification and task assignment to relevant personnel.
- Built a web-based platform using Vue.js and Spring Boot, supporting CRUD operations for road crack data and improving usability for maintenance teams.

Extracurricular Activities & Leadership

Heilongjiang University of Science and Technology AI Association Sep 2021 – Sep 2022
Technical Consultant, Training Group Leader

- Guided and mentored a team of 60+ junior members, focusing on practical skills in machine learning, data analysis, and model optimization.
- Provided technical oversight for team-led AI projects through code reviews, algorithm optimization, and project presentations.
- Fostered a collaborative learning environment by creating a knowledge-sharing platform, encouraging members to discuss challenges and share innovative solutions.

Awards & Distinction

- Excellent Capstone Project (Rank: 1/158), Heilongjiang University of Science and Technology Jun 2024
- Second Prize, The Chinese Mathematics Competition (CMC), China Mathematical Society Mar 2023
- Second Prize, National English Competition for College Students (NECCS), International English Language Teachers Association Oct 2022

- Second Prize, MathorCup National College Mathematical Modeling Challenge Competition, Chinese Society of Optimization, Overall Planning and Economic Mathematics Oct 2022
- Third Prize, Lanqiao Cup National Software and Information Technology Professionals Competition, Talent Exchange Center Of Ministry Of Industry And Information Technology May 2022
- Third Prize, MathorCup National College Mathematical Modeling Challenge Competition – Big Data Competition, Chinese Society of Optimization, Overall Planning and Economic Mathematics Mar 2022
- Third Prize, Northeast Three Provinces Mathematical Modeling League, Northeast Three Provinces Mathematical Modeling League Organizing Committee Aug 2021

Skills & Language

Programming Skills: Python (Advanced), C (Advanced), Java (Intermediate)

Python Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn, SciPy

Professional Applications: MATLAB, LINGO, SPSS, CoppeliaSim

Language: English (Proficient, IELTS 6.5); Mandarin (Native)